

Comprehensive Calculus Examination:

Integration

Instructions: Evaluate each of the following 40 integrals. Show all step-by-step mathematical working clearly. Remember to include the constant of integration (C) for all indefinite integrals.

1. $\int (6x^5 - 3x^2 + 7) dx$
2. $\int \left(\frac{4}{x^3} - \frac{2}{x} \right) dx$
3. $\int \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}} \right) dx$
4. $\int e^{3x} dx$
5. $\int 5^x dx$
6. $\int (2x + 1)^2 dx$
7. $\int \frac{x^3 + 2x^2 + 1}{x^2} dx$
8. $\int \cos(4x) dx$
9. $\int \sec^2(3x) dx$
10. $\int \csc(2x) \cot(2x) dx$
11. $\int (1 + \tan^2(x)) dx$
12. $\int \frac{1}{\sqrt{1-x^2}} dx$
13. $\int \frac{1}{16+x^2} dx$
14. $\int \frac{1}{x\sqrt{x^2-9}} dx$
15. $\int \sinh(5x) dx$
16. $\int \operatorname{sech}^2(x) dx$
17. $\int \tanh(x) dx$
18. $\int \cosh^2(x) dx$
19. $\int \frac{1}{\sqrt{x^2+1}} dx$
20. $\int \frac{1}{\sqrt{x^2-1}} dx$
21. $\int \operatorname{sech}^4(4x) dx$
22. $\int 2x \sin(x^2 + 1) dx$
23. $\int \frac{e^{2x}}{e^{2x}+5} dx$
24. $\int x^2 \sqrt{2x^3 - 4} dx$
25. $\int \frac{[\ln(x)]^3}{x} dx$
26. $\int \sin^4(x) \cos(x) dx$
27. $\int \frac{x+1}{x^2+2x+5} dx$
28. $\int \frac{1}{x \ln(x)} dx$
29. $\int_0^1 3x^2(x^3 + 1)^4 dx$
30. $\int_0^{\pi/2} \cos(x) \sin^3(x) dx$
31. $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$
32. $\int_0^{\ln(3)} e^x \sqrt{e^x + 1} dx$
33. $\int_0^{\pi/4} \sec^2(x) \tan(x) dx$

34. $\int_1^{e^2} \frac{(\ln x)^2}{x} dx$

38. $\int x^2 \ln(x) dx$

35. $\int x e^{2x} dx$

39. $\int x^2 e^{-x} dx$

36. $\int x \cos(3x) dx$

40. $\int e^x \sin(x) dx$

37. $\int \ln(x) dx$

— End of Examination —